Supervised Machine Learning for Multi-Component Condition Classification in Industrial Hydraulic Systems

Sahil Khan Pathan
Roll No. 24AD1199

Vedant Mali
Roll No. 24AD1145

Sejal Zende
Roll No. 24AD1158

Vedika Singh
Roll No. 24AD1147

Department of Artificial Intelligence and Data Science Engineering
DY Patil University, Navi Mumbai, India

***Abstract*—Unplanned failures of hydraulic power units stop the production lines they drive, which makes data-driven condition monitoring an attractive alternative to fixed-interval maintenance. This paper frames the condition monitoring of a hydraulic test rig as four independent supervised multiclass classification problems—cooler efficiency, valve switching behaviour, internal pump leakage and accumulator gas pre-charge pressure—and studies how four classifier families behave on them: multinomial logistic regression, random forest, gradient-boosted trees (XGBoost) and an RBF-kernel support vector machine. Each 60-second operating cycle of 17 sensors (43,680 raw values) is mapped to a fixed-length input vector, and every learned transformation, including feature selection, is fitted inside a single pipeline to prevent train–validation leakage. Using 2,205 cycles, model selection by stratified 5-fold cross-validation on an 80 % training split yields macro-F1 scores between 0.992 and 0.998, and the selected models reach a held-out macro-F1 of 1.000 for the cooler, valve and pump and 0.988 for the accumulator. A blocked cross-validation that holds out contiguous runs of cycles lowers these estimates to 0.995, 0.984, 0.968 and 0.916 respectively, exposing an optimism of up to 7.6 points in randomly split evaluation. The results show that the accumulator is the hardest target, that linear and kernel models are the most sensitive to the input representation, and that temporally aware validation should be reported alongside random-split scores.**

***Index Terms*—predictive maintenance, condition monitoring, hydraulic systems, multiclass classification, random forest, gradient boosting, support vector machine, cross-validation, data leakage**

I. Introduction

Hydraulic power units supply force and motion to presses, injection-moulding machines and mobile equipment. Their components degrade gradually: coolers foul, valves develop switching lag, pumps leak internally and accumulators lose gas pre-charge. Because degradation leaves a signature in routinely measured pressures, flows, temperatures and motor power, it can in principle be recognised by a statistical model long before a functional failure occurs. Machine learning (ML) is now the dominant tool for this kind of predictive maintenance [3].

The public dataset released by Helwig *et al.* [1], [2] records such a rig under systematically varied component conditions, and has become a standard benchmark. Two properties make it demanding from an ML perspective. First, each observation is a multi-rate multivariate time series with far more raw values (43,680) than there are observations (2,205), so a classifier cannot consume it directly. Second, consecutive cycles are recorded seconds apart under identical settings, so observations are not independent and randomly split evaluation can be optimistic [9], [10].

This work treats the problem primarily as a supervised learning and model evaluation study. Its contributions are:

- a leakage-safe learning pipeline in which imputation, scaling, three-stage feature selection and the classifier are fitted jointly, so cross-validation estimates the whole procedure rather than the classifier alone;
- a controlled comparison of four classifier families on three input representations for each of four targets (48 configurations), with hyperparameters held fixed so that differences are attributable to model family and representation;
- a blocked cross-validation analysis that quantifies how much randomly split scores overstate generalisation on this dataset; and
- an error and interpretability analysis of the selected models, deployed behind an inference-only dashboard.

II. Related Work

Helwig *et al.* [1] introduced the rig and classified its component states with linear discriminant analysis on statistical and spectral features, reporting high accuracy for most fault types and lower accuracy for the accumulator. Their work established that hand-crafted cycle statistics carry most of the discriminative information. Subsequent studies on the same data have applied ensemble learners and deep networks, but frequently report only randomly split accuracy.

Among general-purpose tabular learners, random forests [4] and gradient-boosted trees such as XGBoost [5] are strong defaults because they model feature interactions and are insensitive to monotone transformations of the inputs. Support vector machines [6] remain competitive on small, well-scaled datasets. Carvalho *et al.* [3] survey their use in predictive

TABLE I
SENSOR CHANNELS PER OPERATING CYCLE

Channels	Quantity	Rate [Hz]	Samples
PS1–PS6	Pressure [bar]	100	6000
EPS1	Motor power [W]	100	6000
FS1, FS2	Volume flow [l/min]	10	600
TS1–TS4	Temperature [°C]	1	60
VS1	Vibration [mm/s]	1	60
CE, CP, SE	Virtual (efficiency, power)	1	60

TABLE II
PREDICTION TARGETS AND CLASS BALANCE

Target	Classes	Class counts	ρ
Cooler [%]	3 / 20 / 100	732 / 732 / 741	1.01
Valve [%]	73 / 80 / 90 / 100	360 / 360 / 360 / 1125	3.13
Pump leakage	0 / 1 / 2	1221 / 492 / 492	2.48
Accumulator [bar]	90 / 100 / 115 / 130	808 / 399 / 399 / 599	2.03

ρ: imbalance ratio, largest / smallest class.

maintenance and note that random forest is the most frequently applied method in that literature.

Leakage—information about the evaluation data reaching the training procedure—is a recognised source of inflated results [9]. For data with temporal or grouped structure, Roberts *et al.* [10] recommend blocked cross-validation, in which whole blocks of dependent observations are held out together. We apply that recommendation as a diagnostic.

III. Dataset and Problem Formulation

A. Data

The rig repeats a constant 60-s load cycle while 17 channels are logged (Table I): six pressure sensors and motor power at 100 Hz, two flow sensors at 10 Hz, and four temperatures, a vibration channel and three rig-computed virtual quantities at 1 Hz. One cycle therefore holds $7\times6000+2\times600+8\times60 = 43{,}680$ values. The dataset contains $N = 2{,}205$ cycles with no missing values.

B. Targets

Each cycle carries four condition labels (Table II). They are modelled as four independent multiclass problems rather than one joint label, because the joint label space has $3\times4\times3\times4 = 144$ combinations, most of which occur in only 10 cycles (median 10, maximum 230), and because each component is serviced separately. For a target k with class set $\mathcal{C}_k$, we learn $f_k : \mathbb{R}^d \to \mathcal{C}_k$ from pairs $(\mathbf{x}_i, y_i^{(k)})$, $i = 1, \ldots, N$.

A separate stable-flag marks 756 cycles in which static conditions may not have been reached. It describes the rig rather than a component and is kept as metadata. The dataset annotates condition levels, not failure times, so remaining useful life cannot be estimated and is out of scope.

IV. Methodology

A. Input Representation

Because the channels are sampled at different rates, raw sample positions are not aligned in time and cannot serve directly as features. Each cycle is summarised by a fixed-length vector of 626 descriptors in five groups: distribution statistics (mean, quantiles, RMS, skewness, kurtosis), time-domain descriptors (slope, first-difference statistics, peak counts), 1-s rolling-window statistics, a compact FFT summary (dominant frequency, spectral centroid, log-spaced band-energy ratios) and physically motivated cross-channel quantities (pressure differentials, flow imbalance, hydraulic power and electro-hydraulic efficiency). Unsupervised cleaning removes 4 constant, 4 near-constant and 65 exactly duplicated columns, leaving $d = 553$. Three representations are compared:

- *Baseline*: the 17 per-channel cycle means ($d = 17$);
- *Full*: all 553 cleaned descriptors;
- *Selected*: the 553 descriptors passed through the in-pipeline selector of Section IV-C, which outputs 40.

B. Leakage-Safe Pipeline

Every learned step is a stage of one scikit-learn `Pipeline` [7]: median imputation, standardisation $z_j = (x_j - \mu_j)/\sigma_j$, the optional selector and the classifier. Calling `fit` on a training fold therefore re-estimates imputation medians, scaling moments, selection masks and model parameters from that fold only, and cross-validation scores the complete procedure. The only steps fitted on all cycles are the label-free column removals above, which cannot use target information.

C. Feature Selection as Part of the Model

The *Selected* representation applies three filters in sequence, each fitted on training data:

1) *Correlation filter*: scanning columns in order, a column is dropped if $|r| > 0.95$ with any column already kept, which removes near-duplicates ($553 \to$ about 237).
2) *Mutual information*: the top 50 % of columns by estimated $I(X_j; Y)$ are retained [8], a model-free relevance ranking that captures non-linear dependence (about 118 columns).
3) *Embedded tree importance*: a 200-tree random forest ranks the survivors by mean impurity decrease and the top $k = 40$ are passed on. Unlike the univariate filter, this scores each feature in the context of the others.

D. Classifiers

Four model families with different inductive biases are compared. Hyperparameters are fixed (Table III) rather than tuned per target, so that the 48 configurations differ only in model family and representation.

Multinomial logistic regression (LR) models $P(y = c \mid \mathbf{x}) = \exp(\mathbf{w}_c^\top \mathbf{x} + b_c) / \sum_{c'} \exp(\mathbf{w}_{c'}^\top \mathbf{x} + b_{c'})$ with ℓ_2 regularisation. As a linear model it cannot construct interactions, so its score is a direct probe of how linearly separable a representation is.

Random forest (RF) [4] averages $B = 300$ decorrelated decision trees grown on bootstrap samples with random feature subsets.

TABLE III
FIXED CLASSIFIER HYPERPARAMETERS

Model	Settings
LR	ℓ_2, $C = 1$, L-BFGS, 3000 iterations
RF	300 trees, unlimited depth, min. leaf size 1
XGB	300 trees, depth 5, $\eta = 0.1$, subsample 0.9, column sample 0.8, histogram tree method
SVM	RBF, $C = 10$, $\gamma = 1/(d\,\mathrm{Var}[X])$

Gradient boosting (XGB) [5] builds an additive model $F_M(\mathbf{x}) = \sum_{m=1}^{M} \eta\, h_m(\mathbf{x})$ by fitting each tree h_m to the gradient of the regularised multiclass log-loss

$$\mathcal{L} = \sum_i \ell\big(y_i, F(\mathbf{x}_i)\big) + \sum_m \Omega(h_m), \quad \Omega(h) = \gamma T + \tfrac{1}{2}\lambda\|\mathbf{w}\|^2, \tag{1}$$

where Ω penalises the number of leaves T and the leaf weights $\mathbf{w}$.

Support vector machine (SVM) [6] uses the RBF kernel $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x}-\mathbf{x}'\|^2)$ with one-vs-one decomposition for multiple classes.

E. Class Imbalance

When the imbalance ratio ρ exceeds 1.5 (valve, pump, accumulator), each class c receives weight $w_c = N/(|\mathcal{C}|\, N_c)$. LR, RF and SVM use this through their `class_weight` option. Because XGBoost accepts only per-sample weights, a thin wrapper computes w_{y_i} inside `fit` from the labels of the current fold, so weights never depend on validation data. The near-uniform cooler target is left unweighted.

F. Evaluation Protocol and Metrics

For each target, the cycles are split once into 80 % training (1,764) and 20 % test (441), stratified by class. Model selection uses stratified 5-fold cross-validation on the training set only; the configuration with the highest mean CV macro-F1 is selected (ties go to the simpler representation and model), refitted on the full training set, and scored exactly once on the test set.

The headline metric is the macro-averaged F1 score,

$$\mathrm{F1}_{\text{macro}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{2\,P_c R_c}{P_c + R_c}, \tag{2}$$

where P_c and R_c are the precision and recall of class c. It weights every condition level equally, which matters because the rare severe-fault classes are the costly ones. Accuracy, weighted F1 and one-vs-rest ROC-AUC are also reported.

G. Blocked Cross-Validation

Neighbouring cycles share operating settings and slowly drifting temperatures, so a random fold assignment places near-identical cycles on both sides of the split. To measure the effect, the cycle index is partitioned into contiguous blocks of 25 cycles and `StratifiedGroupKFold` (5 folds) is run with the block as the group over all 2,205 cycles, so a validation cycle's immediate neighbours are never in the training folds except at block boundaries. This diagnostic is applied to the selected configuration of each target; the difference between random and blocked CV macro-F1 is reported as the *optimism* of the random split.

H. Implementation

The system is implemented in Python with scikit-learn [7] and XGBoost [5]. All random processes use seed 42. Fitted pipelines are serialised with joblib and served by a Streamlit dashboard that performs inference only.

V. RESULTS

A. Model Comparison

Fig. 1 and Table IV summarise the 48 cross-validated configurations. Three patterns stand out.

Tree ensembles are robust to the representation. On the 17 channel means, RF and XGB already exceed 0.96 macro-F1 for every target, because axis-aligned splits and learned interactions extract non-linear structure that the means leave implicit.

Linear and kernel models depend strongly on it. On the baseline, LR scores 0.672 on the valve and 0.642 on the accumulator, and SVM 0.529 on the valve. With the full representation, LR rises to 0.965 and 0.977; with the selected 40 descriptors LR reaches 0.991 on the valve and SVM 0.989. The information needed to separate these classes is present in the richer representation in a form that is close to linearly separable.

Selection reduces dimension without cost. Moving from 553 to 40 inputs changes the best score by at most 0.002 for any target and improves the kernel and linear models on the valve and pump, which suffer most from irrelevant dimensions. The cooler is solved by every model on every representation (0.997–0.998), and the rule prefers the simplest option, a 17-input XGB model.

Fold-to-fold standard deviations of the selected configurations are 0.002–0.005, smaller than most between-model differences on the valve and accumulator but comparable to the differences among the top three configurations for every target. Rankings within 0.002 should therefore not be over-interpreted.

B. Held-Out Test Performance

Table V reports the single evaluation of each selected model on its 441 test cycles. Cooler, valve and pump are classified without error. The accumulator model reaches 0.989 accuracy and 0.988 macro-F1, with an ROC-AUC of 0.9999, indicating that the remaining errors come from the decision threshold rather than from poor ranking.

C. Generalisation Gap Under Blocked Validation

Fig. 2 and Table VI compare random CV, blocked CV and the test score. The ordering is consistent: blocked CV is lowest for every target, and the gap grows with task difficulty, from 0.003 for the cooler to 0.076 for the accumulator. Since the test set was drawn by the same random procedure as the CV

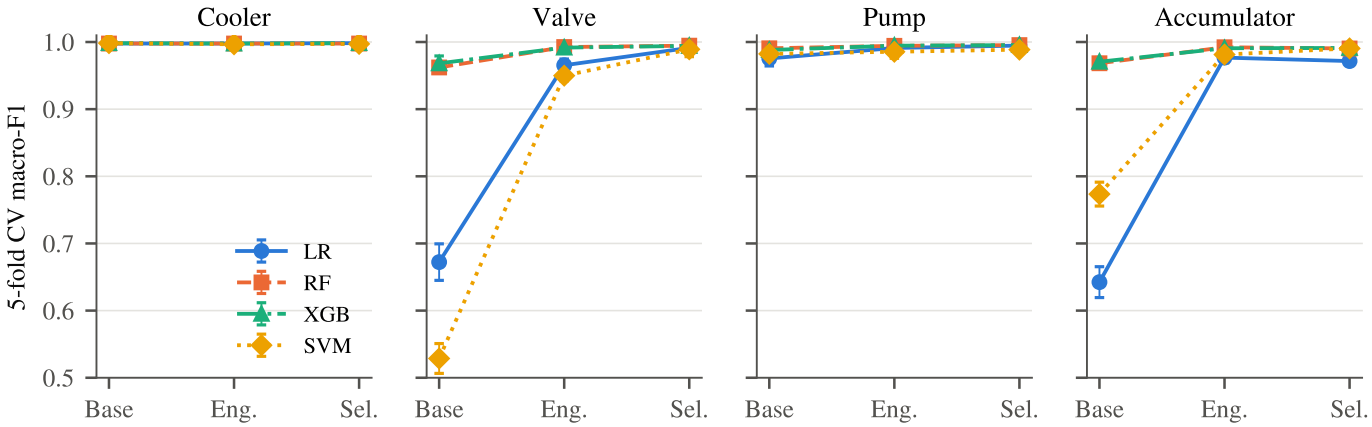


Fig. 1. Stratified 5-fold CV macro-F1 (training set) for each classifier and input representation. Error bars show ± 1 standard deviation across folds. Base: 17 channel means; Eng.: 553 descriptors; Sel.: 40 selected descriptors.

TABLE IV
STRATIFIED 5-FOLD CV MACRO-F1 ON THE TRAINING SET (MEAN OVER FOLDS). BOLD: CONFIGURATION SELECTED FOR EACH TARGET.

Representation	Model	Cooler	Valve	Pump	Accumulator
Baseline ($d = 17$)	LR	0.9977	0.6722	0.9755	0.6424
	RF	0.9977	0.9621	0.9906	0.9684
	XGB	**0.9983**	0.9682	0.9881	0.9706
	SVM	0.9983	0.5287	0.9823	0.7735
Full ($d = 553$)	LR	0.9977	0.9654	0.9911	0.9768
	RF	0.9977	0.9925	0.9943	**0.9922**
	XGB	0.9977	0.9917	0.9946	0.9907
	SVM	0.9966	0.9499	0.9855	0.9813
Selected ($d = 40$)	LR	0.9983	0.9913	0.9947	0.9716
	RF	0.9977	**0.9946**	0.9954	0.9904
	XGB	0.9983	0.9942	**0.9954**	0.9911
	SVM	0.9972	0.9891	0.9886	0.9905

TABLE V
SELECTED MODELS ON THE HELD-OUT TEST SET (441 CYCLES EACH)

Target	Model	Input	Acc.	$F1_m$	AUC
Cooler	XGB	Base (17)	1.000	1.000	1.000
Valve	RF	Sel. (40)	1.000	1.000	1.000
Pump	XGB	Sel. (40)	1.000	1.000	1.000
Accumulator	RF	Full (553)	0.989	0.988	0.9999

TABLE VI
RANDOM VS. BLOCKED 5-FOLD CV OF THE SELECTED MODELS

	Macro-F1		Accuracy	
Target	Random	Blocked	Random	Blocked
Cooler	0.998	0.995	0.998	0.995
Valve	0.995	0.984	0.996	0.987
Pump	0.995	0.968	0.997	0.979
Accumulator	0.992	0.916	0.993	0.933

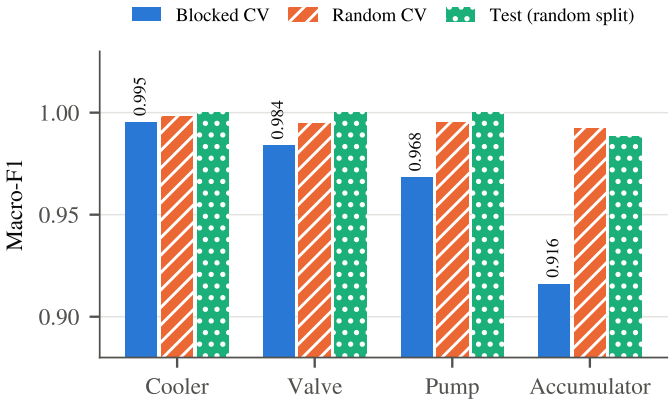


Fig. 2. Macro-F1 of the selected model per target under blocked CV (blocks of 25 consecutive cycles), random stratified CV and the random-split test set. The y-axis starts at 0.88.

folds, it shares their optimism, and the perfect test scores are best read as an upper bound. Blocked CV is the more realistic estimate of performance on cycles recorded at a different time, for example after the rig has been run for a while.

D. Error Analysis

All five accumulator test errors are cycles with a true pre-charge of 100 bar predicted as 90 bar (Fig. 3); no other confusion occurs. The two lowest pre-charge levels are adjacent on the physical scale, and a lower gas pre-charge alters the

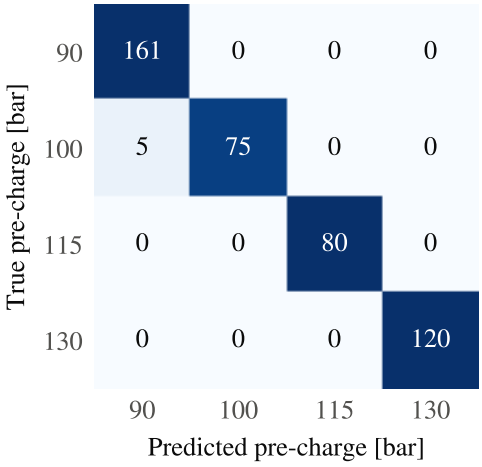


Fig. 3. Test-set confusion matrix of the accumulator model (RF, 553 inputs). Cell shading shows row-normalised recall.

pressure transients only slightly, so these two classes are the most similar in feature space. This agrees with Helwig *et al.* [1], who also reported the lowest classification rates for the accumulator. For maintenance the error is conservative: it overstates degradation and would trigger an earlier inspection, not a missed failure.

E. Interpretability

Table VII lists the inputs with the largest permutation importance (mean drop in test macro-F1 over five random shuffles of one input column). The rankings are physically plausible. The cooler model relies on the virtual cooling power CP and on cooling efficiency, which directly express heat removal. The valve model depends almost entirely on the coefficient of variation of pressure PS2, consistent with switching lag distorting the pressure trace. The pump model uses the upper quartile of flow FS1 and the peak motor power, since internal leakage reduces delivered flow for the same electrical input.

The accumulator model is the exception: no single input has an importance above 0.0011. With 553 highly correlated inputs, the forest spreads its splits across many interchangeable descriptors, so shuffling any one of them is compensated by the others. This is a known limitation of permutation importance under collinearity and not evidence that the model uses no signal; grouped importance or selection on this target would give a clearer picture.

F. Computational Cost

Mean fit times per CV fold are below 0.9 s for all models on the 17-input baseline, and between 0.06 and 9.1 s on the 553-input representation, where XGB is the slowest (9.1 s on the accumulator). The selected pipelines take 1.4–3.1 s, most of it spent in the mutual-information and tree-importance stages. Inference for one cycle, including descriptor extraction, runs in well under a second, which is negligible relative to the 60-s cycle.

TABLE VII
TOP INPUTS BY PERMUTATION IMPORTANCE (TEST SET)

Target	Input	Importance
Cooler	CP mean (cooling power)	0.173
	PS2 mean	0.008
Valve	PS2 coefficient of variation	0.182
	PS2 std. of first difference	0.007
Pump	FS1 75th percentile	0.443
	EPS1 maximum	0.093
Accumulator	PS3 rolling-mean maximum	0.001
	PS1 rolling-std maximum	0.001

VI. Discussion and Limitations

Selection on random CV. Model selection used randomly split CV, while blocked CV was computed only for the winner. A configuration that exploits neighbouring-cycle similarity is favoured by this rule; the accumulator's 553-input forest, which has the largest optimism, may be one. Using blocked CV as the selection criterion is the stricter design and is left to future work.

Single rig, laboratory faults. All data come from one test rig with deliberately set fault levels. Transfer to other machines, gradual in-service degradation and simultaneous unseen faults is not established.

Virtual channels. CE, CP and SE are computed by the rig's own model. The cooler model's reliance on CP means its accuracy partly reflects that model, and would need re-validation where such channels are unavailable.

Fixed hyperparameters. Holding hyperparameters fixed makes the comparison fair but may understate the linear and kernel models, whose defaults are less forgiving than those of tree ensembles.

No remaining useful life. The labels are condition levels without run-to-failure information, so the system classifies current state and does not forecast time to failure.

VII. Conclusion and Future Work

Four classifier families were compared on four hydraulic condition targets inside a leakage-safe pipeline. Tree ensembles are strong on every target and representation, while linear and kernel models need the richer representation to become competitive. The selected models classify cooler, valve and pump without error on the held-out test set and reach 0.988 macro-F1 on the accumulator, whose only confusion is between the two lowest pre-charge levels. Blocked cross-validation lowers these estimates to 0.995, 0.984, 0.968 and 0.916, showing that randomly split scores on this dataset overstate generalisation by up to 7.6 points and should be reported together with a temporally aware estimate.

Future work will select models by blocked CV, add grouped feature importance for the accumulator, explore a single multi-output model that shares representation across targets, and test transfer to data recorded on other hydraulic units.

REFERENCES

[1] N. Helwig, E. Pignanelli, and A. Schütze, "Condition monitoring of a complex hydraulic system using multivariate statistics," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf. (I2MTC)*, Pisa, Italy, 2015, pp. 210–215, doi: 10.1109/I2MTC.2015.7151267.

[2] N. Helwig, E. Pignanelli, and A. Schütze, "Condition monitoring of hydraulic systems," UCI Machine Learning Repository, 2015, doi: 10.24432/C5CW21.

[3] T. P. Carvalho, F. A. A. M. N. Soares, R. Vita, R. P. Francisco, J. P. Basto, and S. G. S. Alcalá, "A systematic literature review of machine learning methods applied to predictive maintenance," *Comput. Ind. Eng.*, vol. 137, Art. no. 106024, 2019.

[4] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

[5] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2016, pp. 785–794.

[6] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995.

[7] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[8] A. Kraskov, H. Stögbauer, and P. Grassberger, "Estimating mutual information," *Phys. Rev. E*, vol. 69, no. 6, Art. no. 066138, 2004.

[9] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Trans. Knowl. Discov. Data*, vol. 6, no. 4, Art. no. 15, 2012.

[10] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.